

parameter is the file name. The second parameter is the *callback* function, which gets called whenever the watched file is touched or edited.

Line 18: Print the file name, size and time modified.

To execute the script at the command prompt, change the file permissions and type the file name:

```
$ chmod +x watch-file.js
$ ./watch-file.js
Syntax: ./watch-file.js <filename>
$ ./watch-file.js watch.txt
Filename being watched: watch.txt
```

In another terminal (2), either *'touch watch.txt'* or edit the *watch.txt* file. After completing the operation, a line gets displayed like the one below at the terminal (1).

```
watch.txt: (size:0) Tue Sep 17 2013 20:01:03 GMT+0630
(India Standard Time)
```

JavaScript is known for its string manipulation and extensive parsing capabilities. The special features of this language, like event driven programming, anonymous functions and closures, make shell programming a lot more fun.

A Web application with JavaScript on the server side

There were many popular scripting languages that were the *de facto* choice at different periods of time. Perl has excellent capabilities for text manipulation and is well suited for CGI scripting. Python and PHP were later preferred for their object oriented nature and embedded scripting into HTML files. So, for complete Web development, a programmer needed another scripting language exclusively for the server side.

Now with Node, that necessity is removed and the Web developer can just do complete application development using JavaScript.

Let's now develop a Web application that generates a quiz questionnaire, reading from a text file as input. The input is a text file where the quiz master can type the quiz questions and multiple choice answers.

Given below is a sample of the input file with the title in the first line and with two records:

```
1 JavaScript quiz
2 Which of the following words are not reserved words in
JavaScript?:break:delete:alpha:false
3 Which of the following are valid data types in JavaScript?
?:Boolean:float:String:Number
```

Here the questions and answers are separated by "..."

and multiple choice answers are separated by "...".

Given below is the complete code for the JavaScript quiz..

```
1 var http = require('http');
2 var fs = require('fs');
3 var question = "";
4 var options_array = "";
5 var quiz_heading = "";
6 var data_lines = {};
7
8 fs.readFile('./questions.txt', 'utf8',
    function (err, data) {
9     if (err) { return console.log(err); }
10    else
11    {
12        data_lines = data.split('\n');
13    }
14 });
15
16 var server = http.createServer(function(req, res) {
17     res.writeHead(200, {'Content-Type': 'text/html'});
18     res.write('<html><head><title>'+ quiz_heading
19             + '</title></head>');
20     res.write('<body>');
21
22     //Extract QUIZ heading, which is the first line
23     quiz_heading = data_lines[0];
24     res.write('<h1 align="center">'+quiz_heading+'</
h1>');
25
26     res.write('<form>');
27     for (var i=1; i < data_lines.length; i++)
28     {
29         if (data_lines[i].length > 1)
30         {
31             var q_and_a = data_lines[i].split(":");
32             question = q_and_a[0];
33             options_array = q_and_a[1].split(';');
34
35             res.write('<h4>'+ i + ' ' + question+'<
h4>');
36
37             for (j = 0; j < options_array.length; j++)
38             {
39                 res.write('<input type="checkbox">'
40                         + options_array[j]+'</
br>');
41             }
42             res.write('<br><br><input type="submit"
43                     value="Evaluate">');
44             res.write('</form></body></html>');
```